

LOCAL COMPONENT GROUPS AND THE ℓ -ADIC VALUATIONS OF THE MODULAR DEGREE

[AUTHOR]

ABSTRACT. Let $E/\mathbb{Q}$ be the $X_0(N)$ -optimal curve of its isogeny class and $\varphi_E: X_0(N) \rightarrow E$ its modular parametrisation. For a bad prime q let Φ_q^{geom} denote the component group of the Néron model of E over $\overline{\mathbb{F}}_q$. We study the inequality

$$v_\ell(\deg \varphi_E) \geq \sum_{q|N} v_\ell(\#\Phi_q^{\text{geom}})$$

for odd primes ℓ at which $E[\ell]$ is irreducible, together with its refinement in which, at each odd q , the component group of E is replaced by the larger of the component groups of E and of its quadratic twist by $\mathbb{Q}(\sqrt{q^*})$. For $\ell \geq 5$ the multiplicative part is a consequence of a theorem of Kim and Ota on the Pollack–Weston conjecture combined with a result of Agashe, Ribet and Stein, under hypotheses that exclude $\ell = 3$, primes $\ell \mid N$ and primes $q \equiv \pm 1 \pmod{\ell}$. We verify both inequalities with no exception on every optimal curve of conductor at most 500000 (more than two million curves) and on 893 curves of conductor up to $4 \cdot 10^6$ outside the tables, beyond all of these hypotheses. The new feature is the additive primes: at $\ell = 3$ every prime of Kodaira type IV or IV* and every odd prime of type II or II* contributes one, whatever its Tamagawa number, and at every odd ℓ an odd prime of type I $_n^*$ contributes $v_\ell(n)$. We prove the qualitative statements by level lowering, applied to a quadratic twist in the twisted cases, formulate the quantitative conjecture, and describe the exact failure of the inequality in the Eisenstein case.

1. INTRODUCTION

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N and let $f_E \in S_2(\Gamma_0(N))$ be the newform attached to it. Among the curves in the isogeny class of E there is a distinguished one, the $X_0(N)$ -optimal curve, for which the modular parametrisation $\varphi_E: X_0(N) \rightarrow E$ has minimal degree; throughout, E denotes this curve and $\deg \varphi_E$ its *modular degree*. The modular degree is computable [7, 24] and its growth is understood (it is $N^{1+o(1)}$ times a power of the discriminant), but its prime factorisation is not. The purpose of this note is to describe, and partly prove, a local-to-global divisibility: the ℓ -adic valuation of $\deg \varphi_E$ is bounded below by local data at the bad primes.

For a bad prime q let $\Phi_q^{\text{geom}} = \Phi_q(\overline{\mathbb{F}}_q)$ be the group of components of the special fibre of the Néron model of E at q , over the algebraic closure. Its order is read off the Kodaira type: n for type I $_n$; 1 for II, II*; 2 for III, III*; 3 for IV, IV*; 4 for I $_0^*$ and I $_n^*$ [20, IV.9]. The Tamagawa number $c_q = \#\Phi_q(\mathbb{F}_q)$ counts only the components defined over $\mathbb{F}_q$; the statements below are about Φ_q^{geom} , not c_q .

Conjecture 1.1. *Let E be $X_0(N)$ -optimal and without complex multiplication, and let ℓ be an odd prime such that $E[\ell]$ is irreducible as a $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ -module. Then*

$$(1) \quad v_\ell(\deg \varphi_E) \geq \sum_{q|N} v_\ell(\#\Phi_q^{\text{geom}}).$$

For $\ell \geq 5$ only multiplicative primes contribute, through $v_\ell(n_q)$ where $n_q = v_q(\Delta_E)$ is the exponent of the minimal discriminant. For $\ell = 3$ the primes of type IV and IV* contribute

one each. The multiplicative part has a long history. Ribet and Takahashi [19, 22] compared $\deg \varphi_E$ with the degree of a parametrisation by a Shimura curve and found the product of the n_q over the primes of the quaternion discriminant, up to primes where $E[\ell]$ is reducible; Pasten [14] controls that error term. Pollack and Weston [15] conjectured, and Kim and Ota [11] proved, that moving a prime q into the “new” part of the level changes the congruence ideal of f_E by exactly the *Tamagawa exponent* t_q , the largest t with $\rho_{E,\ell} \bmod \ell^t$ unramified at q ; for multiplicative q one has $t_q = v_\ell(n_q)$ whether or not the reduction is split. Together with the theorem of Agashe, Ribet and Stein [1] that $\deg \varphi_E$ divides the congruence number r_E , with $v_p(\deg \varphi_E) = v_p(r_E)$ whenever $p^2 \nmid N$, this gives:

Theorem 1.2 (Kim–Ota, Agashe–Ribet–Stein). *Let $\ell \geq 5$ be a prime not dividing N such that $\bar{\rho}_{E,\ell}$ is absolutely irreducible when restricted to $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, $\ell^* = (-1)^{(\ell-1)/2}\ell$. Then*

$$v_\ell(\deg \varphi_E) \geq \sum_{\substack{q \parallel N \\ q \not\equiv \pm 1 \pmod{\ell}}} v_\ell(n_q).$$

The data presented in Section 4 say that none of the three restrictions in Theorem 1.2 is needed for the inequality: it holds at $\ell = 3$, at primes ℓ dividing N , and with the primes $q \equiv \pm 1 \pmod{\ell}$ included, in every one of more than two million optimal curves.

The additive primes enter in a second way, through quadratic twists. For an odd prime q put $q^* = (-1)^{(q-1)/2}q$ and let $E^{(q)}$ be the quadratic twist of E by $\mathbb{Q}(\sqrt{q^*})$. The twisting character is ramified only at q , so $E^{(q)}$ has the same reduction type as E at every prime other than q , while at q the Kodaira types are exchanged (Lemma 2.4):

$$I_n \leftrightarrow I_n^*, \quad \text{II} \leftrightarrow \text{IV}^*, \quad \text{IV} \leftrightarrow \text{II}^*, \quad \text{III} \leftrightarrow \text{III}^*, \quad I_0 \leftrightarrow I_0^*$$

(the first for every odd q , the others for $q \geq 5$). For an odd prime ℓ and a bad prime q define the *twist-symmetrised* weight

$$w_\ell(E, q) = \begin{cases} v_\ell(n) & \text{if the type at } q \text{ is } I_n, \text{ or } I_n^* \text{ with } q \text{ odd,} \\ 1 & \text{if } \ell = 3 \text{ and the type is IV or IV}^*, \text{ or II or II}^* \text{ with } q \text{ odd,} \\ 0 & \text{otherwise,} \end{cases}$$

where n is the index of the Kodaira symbol. For $q \geq 5$ this is the twist-symmetrised component order,

$$w_\ell(E, q) = \max(v_\ell(\#\Phi_q^{\text{geom}}(E)), v_\ell(\#\Phi_q^{\text{geom}}(E^{(q)}))),$$

and for $q = 2$ it is just $v_\ell(\#\Phi_2^{\text{geom}}(E))$; so $w_\ell(E, q) \geq v_\ell(\#\Phi_q^{\text{geom}}(E))$ always, with equality unless q is odd and of type II, II^{*} or I_n^{*}, the *twisted* types.

Conjecture 1.3. *With the hypotheses of Conjecture 1.1,*

$$(2) \quad v_\ell(\deg \varphi_E) \geq \sum_{q \mid N} w_\ell(E, q).$$

Conjecture 1.3 implies Conjecture 1.1. We keep both because they have different standing: (1) is a statement about E alone, with Theorem 1.2 behind its multiplicative part, whereas (2) says that an odd prime of type II forces a factor 3 which its component group of order one does not see, and that the prime 2 does *not* do so (Section 4.5). The qualitative half of the additive terms is a theorem:

Theorem 1.4. *Let $E/\mathbb{Q}$ have conductor N and let ℓ be an odd prime with $\ell^2 \nmid N$ such that $\bar{\rho}_{E,\ell}$ is irreducible and, if $\ell = 3$, absolutely irreducible when restricted to $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$. Let $q \neq \ell$ be a prime at which E has additive reduction of one of the following kinds:*

- (a) type IV or IV*, with $q \geq 5$ and $\ell = 3$;
- (b) type II or II*, with $q \geq 5$ and $\ell = 3$;
- (c) type I_n^* with $\ell \mid n$, q odd.

Then ℓ divides the modular degree of the $X_0(N)$ -optimal curve in the isogeny class of E .

In each case the prime q has $w_\ell(E, q) \geq 1$, and the theorem says that it contributes at least one to $v_\ell(\deg \varphi_E)$. The quantitative half, namely that the contributions of several such primes and of the multiplicative primes add up and that a prime of type I_n^* contributes the full $v_\ell(n)$, is Conjecture 1.3; the data support it also at $q = 3$ and when $\ell^2 \mid N$, where the proof does not apply. In case (a) the mechanism is that the ramification of $\bar{\rho}_{E,3}$ at q is unipotent, so that the level can be lowered from q^2 to q (Lemma 2.2). In cases (b) and (c) $\bar{\rho}_{E,\ell}$ itself has conductor exponent 2 at q and nothing can be lowered; but its twist $\bar{\rho}_{E,\ell} \otimes \chi_q$, the representation of $E^{(q)}$, has conductor exponent 1 or 0 at q . Lowering the level of the twist and twisting back produces an eigenform of level dividing N , different from f_E and congruent to it modulo ℓ (Section 3). The argument does not apply at $q = 2$, where a quadratic character ramified at 2 changes the conductor exponent, and the data say that the twisted types at 2 indeed force nothing.

The third statement concerns the Eisenstein case, in which Conjecture 1.1 can fail.

Conjecture 1.5. *Let E be optimal, not CM, with a rational ℓ -isogeny, ℓ odd, and let D be the deficit $\sum_q v_\ell(\#\Phi_q^{\text{geom}}) - v_\ell(\deg \varphi_E)$ of (1). Then*

$$D \leq v_\ell(\#E(\mathbb{Q})_{\text{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3],$$

and also $D \leq$ the number of curves in the isogeny class of E with a rational point of order ℓ . In particular $D \leq 0$ when E has no rational ℓ -torsion point, $D \leq 1$ for $\ell \geq 5$, and $D \leq 2$ for $\ell = 3$.

Section 2 collects the local facts, Section 3 proves Theorem 1.4, Section 4 describes the computations and Section 5 discusses the prime 2, where the analogous statements overlap with the literature on Watkins's conjecture.

2. LOCAL INVARIANTS

Let q be a prime of bad reduction. We use Tate's algorithm [20, IV.9] for the Kodaira type and c_q , and the following facts.

Lemma 2.1. *Let q be a prime of multiplicative reduction, $n_q = v_q(\Delta_E)$, and $\ell \neq q$ a prime. Then $\rho_{E,\ell} \bmod \ell^t$ is unramified at q if and only if $\ell^t \mid n_q$. This holds for split and for nonsplit reduction.*

Proof. Over the unramified quadratic extension of $\mathbb{Q}_q$ (which does not change the inertia group) E is a Tate curve with parameter u , $v_q(u) = n_q$. The ℓ^t -torsion is generated by μ_{ℓ^t} and an ℓ^t -th root of u , so inertia acts trivially on $E[\ell^t]$ exactly when u is an ℓ^t -th power in $\mathbb{Q}_q^{\text{ur}}$ up to units, that is, when $\ell^t \mid n_q$. $\square$

Lemma 2.2. *Let $q \geq 5$ be a prime at which E has reduction of type IV or IV*. Then the image of the inertia group I_q in $\text{Aut}(T_3 E)$ is cyclic of order 3, and its image in $\text{Aut}(E[3])$ is a nontrivial unipotent subgroup. Consequently $E[3]^{\ell^t}$ is a line, $\bar{\rho}_{E,3}$ is tamely ramified at q , and the Serre conductor of $\bar{\rho}_{E,3}$ has exponent 1 at q .*

Proof. For $q \geq 5$ the reduction is potentially good and the semistability defect, the order of the image of I_q in $\text{Aut}(T_\ell E)$ for any $\ell \neq q$, equals $12/\gcd(v_q(\Delta_E), 12)$ [12]; for $v_q(\Delta_E) \in \{4, 8\}$ this is 3. The image is tame since $q \neq 3$, hence cyclic, generated by an element σ of order 3 with determinant 1, so with eigenvalues the two primitive cube roots of unity. Hence $\sigma^2 + \sigma + 1$ annihilates $T = T_3 E$, and T is a module over $\mathbb{Z}_3[\sigma]/(\sigma^2 + \sigma + 1) \simeq \mathbb{Z}_3[\zeta_3]$, a discrete valuation ring

with uniformiser $\pi = 1 - \zeta_3$ and residue field $\mathbb{F}_3$. Being torsion-free of $\mathbb{Z}_3$ -rank 2, T is free of rank one over $\mathbb{Z}_3[\zeta_3]$, and $E[3] = T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$ with σ acting as multiplication by $\zeta_3 = 1 - \pi$. Since $\pi \neq 0$ in $\mathbb{Z}_3[\zeta_3]/(\pi^2)$, this is a nontrivial unipotent automorphism of order 3; its fixed space is the line $\pi\mathbb{Z}_3[\zeta_3]/(\pi^2)$. The Serre conductor exponent at q is $\dim E[3] - \dim E[3]^{I_q} = 1$, there being no wild part. $\square$

Remark 2.3. For the other additive types at $q \geq 5$ the image of inertia mod 3 is not unipotent: it has order 2 or 4 (types I_0^* , III, III^* , defects 2 and 4), or it is minus a unipotent (types II, II^* , defect 6), or it is a ramified quadratic character times a unipotent (type I_n^*). In the last two cases $E[3]^{I_q} = 0$ and the Serre conductor of $\bar{\rho}_{E,3}$ has exponent 2 at q , the same as the conductor of E ; these are the twisted types of Conjecture 1.3.

Lemma 2.4. *Let q be an odd prime, $\chi = \chi_q$ the quadratic character of conductor q and $E^{(q)}$ the twist of E by χ , that is, by $\mathbb{Q}(\sqrt{q^*})$.*

- (i) *At every prime $p \neq q$ the curves E and $E^{(q)}$ have the same Kodaira type, the same geometric component group and the same conductor exponent.*
- (ii) *If E has type I_n at q ($n \geq 0$) then $E^{(q)}$ has type I_n^* at q , and conversely, with the same n . If $q \geq 5$ and E has type II, III, IV at q then $E^{(q)}$ has type IV^* , III^* , II^* respectively, and conversely. The conductor exponent of $E^{(q)}$ at q is 2 when its reduction is additive and 1 when it is of type I_n , $n \geq 1$.*
- (iii) *(Atkin–Li [2]) Let g be a newform of weight 2 on $\Gamma_0(M)$ with $q^2 \nmid M$. Then the twist $g \otimes \chi$, with coefficients $a_n(g \otimes \chi) = \chi(n)a_n(g)$, is a newform of weight 2 on $\Gamma_0(M')$, $M' = \text{lcm}(M, q^2)$; that is, $M' = Mq^2$ if $q \nmid M$ and $M' = Mq$ if $q \parallel M$.*

Proof. (i) The character χ is unramified at p , so E and $E^{(q)}$ become isomorphic over an unramified extension of $\mathbb{Q}_p$, and the Kodaira type, the geometric component group and the conductor exponent are invariant under unramified base change. (ii) If $(a_1, a_2, a_3, a_4, a_6)$ is a minimal model of E at q with invariants c_4, c_6, Δ , the twist has a model with invariants $q^2 c_4, q^3 c_6, q^6 \Delta$ and the same j -invariant. For potentially multiplicative reduction, $v_q(j) < 0$, the type is I_n or I_n^* with $n = -v_q(j)$ according as the reduction is multiplicative or additive, and a ramified quadratic twist exchanges the two. For $q \geq 5$ and potentially good reduction the type is determined by $v_q(\Delta)$ of a minimal model, which lies in $\{0, 2, 3, 4, 6, 8, 9, 10\}$ for the types $I_0, II, III, IV, I_0^*, IV^*, III^*, II^*$; the twisted model has $v_q(\Delta)$ increased by 6, and is minimal exactly when the result is less than 12, otherwise 12 is subtracted. The conductor exponents are the standard ones for q odd (tame). (iii) is [2, Theorem 3.1]; in the case $q \parallel M$, the local component of g at q is an unramified twist of the Steinberg representation, whose twist by a ramified quadratic character has conductor exponent 2. $\square$

3. PROOF OF THEOREM 1.4

Let $f = f_E$, a newform of weight 2 on $\Gamma_0(N)$, let E_0 be the optimal curve in the class of E and $r = r_{E_0}$ its congruence number. By [1, Theorem 2.1], $\deg \varphi_{E_0}$ divides r with $v_p(\deg \varphi_{E_0}) = v_p(r)$ for every prime p with $p^2 \nmid N$; as $\ell^2 \nmid N$ it suffices to prove $\ell \mid r$. For that it is enough to produce a normalised eigenform $h \in S_2(\Gamma_0(N)) \otimes \overline{\mathbb{Q}}$, $h \neq f$, with $a_p(h) \equiv a_p(f) \pmod{\lambda}$ for all primes p outside a finite set, for some prime $\lambda \mid \ell$ of a number field containing the coefficients: such a congruence means that the Hecke algebra of level N has a maximal ideal of residue characteristic ℓ containing the kernels of both eigencharacters, which is equivalent to $\ell \mid r$ [16], [1, §3]. Write $\bar{\rho} = \bar{\rho}_{E,\ell}$, irreducible by hypothesis.

We use the level-lowering theorem in the following form [17, 5, 8] (see [18] for an exposition): if $\bar{\rho}$ arises from a newform of weight 2 on $\Gamma_0(N)$, satisfies the irreducibility hypothesis of the theorem, and $q \neq \ell$ is a prime with $v_q(N(\bar{\rho})) < v_q(N)$, where $N(\bar{\rho})$ is the Serre conductor, then

$\bar{\rho}$ arises from a newform of weight 2 on $\Gamma_0(M)$ for some M dividing N/q . [[Check the exact hypotheses of the statements used, in particular when $\ell \mid N$ and at $\ell = 3$; the condition over $\mathbb{Q}(\sqrt{-3})$ is the one needed in [8].]]

Case (a). Here $q^2 \parallel N$ and $\ell = 3$. By Lemma 2.2 the Serre conductor $N(\bar{\rho})$ has exponent 1 at q , so there is a newform g of weight 2 on $\Gamma_0(M)$, $M \mid N/q$, with coefficients in the ring of integers $\mathcal{O}$ of a number field and a prime $\lambda \mid 3$ of $\mathcal{O}$ such that $a_p(g) \equiv a_p(f) \pmod{\lambda}$ for all primes $p \nmid 3N$. Since $M < N$, the form g is old at level N and lies in the orthogonal complement of f in $S_2(\Gamma_0(N))$; take $h = g$.

Case (b). Here $q^2 \parallel N$, $\ell = 3$, and E has type II or II* at $q \geq 5$. Let $\chi = \chi_q$ and $E' = E^{(q)}$. By Lemma 2.4, E' has type IV* or IV at q , the same types as E elsewhere, and conductor N ; its newform is $f' = f \otimes \chi$, a newform of weight 2 on $\Gamma_0(N)$, and $\bar{\rho}_{E',3} = \bar{\rho} \otimes \chi$ satisfies the irreducibility hypothesis since $\bar{\rho}$ does. By Lemma 2.2 applied to E' , the Serre conductor of $\bar{\rho} \otimes \chi$ has exponent 1 at q , so there is a newform g of weight 2 on $\Gamma_0(M)$, $M \mid N/q$, with $a_p(g) \equiv a_p(f') \pmod{\lambda}$ for all $p \nmid 3N$. As $q^2 \nmid M$, Lemma 2.4(iii) says that $h = g \otimes \chi$ is a newform on $\Gamma_0(M')$ with $M' = \text{lcm}(M, q^2)$, which divides N because $M \mid N/q$ and $q^2 \parallel N$. For $p \nmid 3N$,

$$a_p(h) = \chi(p)a_p(g) \equiv \chi(p)a_p(f \otimes \chi) = \chi(p)^2 a_p(f) = a_p(f) \pmod{\lambda}.$$

Finally $h \neq f$: otherwise $g = h \otimes \chi = f \otimes \chi = f'$ would have level N , whereas $M < N$. So h is an eigenform of level dividing N , different from f and congruent to it, and $3 \mid r$.

Case (c). Here E has type I_n^* at an odd prime $q \neq \ell$ with $\ell \mid n$, so $q^2 \parallel N$. Let $\chi = \chi_q$ and $E' = E^{(q)}$. By Lemma 2.4, E' has multiplicative reduction of type I_n at q and conductor N/q , so $f' = f \otimes \chi$ is a newform of weight 2 on $\Gamma_0(N/q)$ with $q \parallel N/q$. By Lemma 2.1 applied to E' , $\bar{\rho} \otimes \chi = \bar{\rho}_{E',\ell}$ is unramified at q , so by the level-lowering theorem (here in Ribet's original form [17], q exactly dividing the level) there is a newform g of weight 2 on $\Gamma_0(M)$, $M \mid N/q^2$, with $a_p(g) \equiv a_p(f') \pmod{\lambda}$ for all $p \nmid \ell N$. As $q \nmid M$, $h = g \otimes \chi$ is a newform on $\Gamma_0(Mq^2)$ with $Mq^2 \mid N$, congruent to f at all $p \nmid \ell N$ by the computation of case (b), and $h \neq f$ since $g = h \otimes \chi$ would otherwise equal f' , of level $N/q \nmid N/q^2$. Hence $\ell \mid r$. $\square$

Example 3.1. The congruences produced by the proof can be seen directly on the tables (labels as in [7]). (b): the curve 121a1 has type II at 11; its twist by -11 lies in the class 121c and has type IV* at 11; its mod 3 representation has conductor 11 and is congruent to the form of 11a1; twisting back, $f_{11a} \otimes \chi_{-11}$ is the form of 121d, and indeed $a_p(121a1) \equiv \chi_{-11}(p) a_p(11a1) \pmod{3}$ for all $p \neq 3, 11$ (checked for $p < 500$), while $\deg \varphi_{121a1} = 6$. (c) at $\ell = 3$: 539c1 has type I_3^* at 7; its twist by -7 has conductor 77 and type I_3 at 7, and $a_p(539c1) \equiv \chi_{-7}(p) a_p(11a1) \pmod{3}$ for all $p \neq 3, 7, 11$ (checked for $p < 1000$); $\deg \varphi_{539c1} = 2^5 \cdot 3^2$. (c) at $\ell = 5$: 980i1 has type I_5^* at 7; its twist by -7 has conductor 140 and type I_5 at 7, and $a_p(980i1) \equiv \chi_{-7}(p) a_p(20a1) \pmod{5}$ for all $p \neq 2, 5, 7$ (checked for $p < 1000$); $\deg \varphi_{980i1} = 2^6 \cdot 3^2 \cdot 5$.

Remark 3.2. The hypothesis $\ell^2 \nmid N$ is used only to pass from the congruence number to the modular degree, and it cannot simply be dropped from that step: Agashe, Ribet and Stein [1, Proposition 2.3] note that for the curve 99a1 the prime 3 divides the congruence number but not the modular degree. The data nevertheless show the conclusion of the theorem for every curve with $9 \mid N$ and a prime of one of the three kinds; so do they at $q = 3$ in cases (a) and (b), where Lemma 2.2 does not apply because the ramification is wild. The proof gives one factor of ℓ for each such prime but says nothing about additivity with the multiplicative contributions or with each other, nor about the full $v_\ell(n)$ in case (c); that is the content of Conjecture 1.3. Note also that the proof of case (c) is the proof of the multiplicative case at level N/q transported by the twist: the quantitative statement for I_n^* should therefore follow from the Kim–Ota formula for $f \otimes \chi$ once the congruence ideals at levels N/q and N are compared.

4. THE COMPUTATIONS

4.1. Data and protocol. We used Cremona’s tables [7] in the form of the `ecdata` repository at commit `25cec5e`, which contain every elliptic curve over $\mathbb{Q}$ of conductor below 500000 with its Kodaira symbols, Tamagawa numbers, modular degree, isogeny class, torsion and, up to conductor 400000, the images of the mod ℓ Galois representations. Optimality is determined in the tables for every class of conductor at most 400000 and for 359,009 further classes up to 500000. Conjectures 1.1 and 1.5 were found on the curves of conductor at most 300000, written down, and then tested once on the curves of conductor in $(300000, 400000]$, which had been held out, and afterwards on the extension to 500000; nothing was adjusted after the hold-out test. Conjecture 1.3 replaced an earlier, weaker formulation of the same phenomenon (every prime of a twisted type, the prime 2 included, contributing “all but the largest” of the weights), which had passed the hold-out as well. The replacement was formulated on the working set, after the prime 2 had been separated from the odd primes, and then tested once on the hold-out and on the extension (Section 4.5); the earlier formulation is implied by it. The Kodaira types and discriminant exponents at the bad primes were recomputed with PARI/GP 2.15 [23] from the a -invariants, and fifteen modular degrees spanning the configurations were recomputed with `ellmoddegree`, all agreeing with the tables.

4.2. Results. Table 1 gives, for each range and each small prime ℓ , the number of optimal non-CM curves with $E[\ell]$ irreducible and a positive right-hand side in (1) and in (2), the number of violations of each inequality, and the number of Eisenstein curves with a positive bound together with the violations of Conjecture 1.5. There are none.

set	ℓ	curves	(1) > 0	(2) > 0	viol. (1)	viol. (2)	Eis. > 0 / viol.
working set, $N \leq 300000$	3	1,311,066	598,833	705,563	0	0	67,845 / 0
	5	1,311,066	235,510	262,531	0	0	4,454 / 0
	7	1,311,066	121,947	136,707	0	0	759 / 0
	11	1,311,066	42,334	47,040	0	0	0 / 0
	13	1,311,066	27,215	29,943	0	0	26 / 0
hold-out, $300000 < N \leq 400000$	3	429,936	206,749	242,691	0	0	18,225 / 0
	5	429,936	79,550	89,164	0	0	989 / 0
	7	429,936	41,387	46,767	0	0	137 / 0
	11	429,936	14,244	15,970	0	0	0 / 0
	13	429,936	9,212	10,280	0	0	2 / 0
extension, $400000 < N \leq 500000$	3	359,009	181,075	214,033	0	0	9,894 / 0
	5	359,009	70,514	78,650	0	0	474 / 0
	7	359,009	36,734	41,317	0	0	70 / 0
	11	359,009	13,347	14,975	0	0	0 / 0
	13	359,009	8,467	9,465	0	0	5 / 0
out of table, $500000 < N \leq 4 \cdot 10^6$	3	893	561	718	0	0	0 / 0
	5	893	6	7	0	0	0 / 0
	7	893	0	1	0	0	0 / 0

TABLE 1. Verification counts. “(1) > 0” is the number of curves with $E[\ell]$ irreducible, not CM, for which the right-hand side of (1) is positive, and “viol.” the number of violations; likewise for (2). The last column counts the Eisenstein curves with a positive bound in (1) and the violations of Conjecture 1.5.

The theorem range of Theorem 1.2 covers 281,739 of the 913,244 working-set pairs (E, ℓ) with $E[\ell]$ irreducible and a positive multiplicative bound; the remaining pairs have $\ell = 3$ (455,707), $\ell \mid N$ (175,798 at $\ell \geq 5$), or a contributing prime $q \equiv \pm 1 \pmod{\ell}$ (410,074), and show no violation either.

4.3. Isolated configurations. Table 2 isolates the additive types at $\ell = 3$. For the untwisted types it lists, for working-set curves with no multiplicative prime q with $3 \mid n_q$ and all additive primes of a single Kodaira type, the minimum of $v_3(\deg \varphi_E)$ by type and number of primes. For the twisted types it lists the minimum of $v_3(\deg \varphi_E)$ over the curves whose only contribution to (2) comes from primes of the given type at odd q , or at $q = 2$, with the given total weight. The hold-out and the extension reproduce every minimum.

type	weight 1	weight 2	weight 3
IV (any q)	1 (25,414)	2 (969)	3 (11)
IV* (any q)	1 (30,168)	2 (2,268)	3 (45)
III (any q)	0 (34,904)	0 (2,208)	0 (34)
III* (any q)	0 (25,574)	0 (1,307)	0 (29)
I ₀ * (any q)	0 (42,924)	0 (3,093)	0 (53)
II, II* at odd q	1 (56,509)	2 (2,502)	3 (12)
I _n * at odd q , weight $v_3(n)$	1 (28,175)	2 (3,453)	3 (89)
II, II* at $q = 2$	0 (44,672)	—	—
I _n * at $q = 2$, weight $v_3(n)$	0 (21,176)	1 (5,329)	1 (342)

TABLE 2. Minimum of $v_3(\deg \varphi_E)$ (number of curves) over working-set curves with $E[3]$ irreducible whose only possible contributions at 3 come from the primes named in the row, with the given number of primes (untwisted rows) or total weight (twisted rows).

The rows IV, IV* are Conjecture 1.1: each prime contributes exactly one at the minimum, while III, III* and I₀* contribute nothing. The twisted rows are Conjecture 1.3: at an odd prime, each II or II* contributes one and each I_n* contributes $v_3(n)$, additively, and the minimum is attained in every cell; at the prime 2 a type II or II* contributes nothing ($v_3(\deg \varphi_E) = 0$ for 14,182 of the 44,672 curves whose only candidate is such a prime), and a type I_n* with $3 \parallel n$ contributes nothing either, although with $9 \mid n$ the minimum is 1.

4.4. Sharpness and the rest of the degree. The bounds are attained. Among working-set curves with $E[\ell]$ irreducible and a positive bound, $v_\ell(\deg \varphi_E)$ equals the right-hand side of (2) in 39.4% of cases at $\ell = 3$, 71.1% at $\ell = 5$, 81.2% at $\ell = 7$ and 91.5% at $\ell = 13$ (for (1): 36.9%, 69.7%, 80.4% and 91.2%), the proportion decreasing slowly with the conductor (from 77% to 38% at $\ell = 3$ between conductors 1000 and 300000). The whole odd part of $\deg \varphi_E$ is another matter: for the 1,226,216 working-set optimal non-CM curves with no isogeny of odd degree, the odd part of $\deg \varphi_E$ equals $\prod_\ell \ell^{b_\ell}$, with b_ℓ the right-hand side of (2), in only 1.85% of cases. The quotient is most often 9, 3, 15, 45, 27 or 21; its smallest prime factor is 3 in 63% of the cases and divides N in 52%. So the local component groups account for a definite, sharp part of the modular degree and for little of the rest, which is governed by congruences of other origins and by the size of the Petersson norm.

4.5. The twisted primes. Table 3 tests (2) on the three ranges and every odd $\ell \leq 13$, and shows what happens when the twisted types at 2 are counted as well: the inequality then fails thousands of times in each range. The earlier “all but the largest” rule was an artifact of this mixture, since in the data the largest weight usually sat at the prime 2.

Two further tests concern the mechanism. First, for the 56,509 working-set curves whose only candidate contribution at 3 is a single odd prime of type II or II*, 3 divides $\deg \varphi_E$ in every case, uniformly in $q \bmod 3$, in the type, in the Tamagawa number of the twist at q (which is 1 or 3 for the types IV*, IV) and in the image of $\bar{\rho}_{E,3}$; this is Theorem 1.4(b) for $q \geq 5$, and the data extend it to $q = 3$ (26,649 of the curves) and to $9 \mid N$. Second, for the 2,502 working-set curves

set	ℓ	odd twisted weight > 0	viol. (2)	twisted prime 2	viol. if 2 counted
working set, $N \leq 300000$	3	171,272	0	133,473	42,100
	5	31,002	0	25,960	18,955
	7	15,866	0	16,255	13,454
	11	4,900	0	7,331	6,803
	13	2,790	0	5,161	4,891
hold-out, $300000 < N \leq 400000$	3	60,688	0	45,675	13,219
	5	11,147	0	9,371	6,648
	7	5,856	0	5,847	4,746
	11	1,803	0	2,685	2,399
	13	1,095	0	1,906	1,799
extension, $400000 < N \leq 500000$	3	56,703	0	40,415	11,611
	5	9,473	0	7,675	5,406
	7	4,978	0	4,781	3,818
	11	1,679	0	2,183	1,998
	13	1,022	0	1,526	1,383

TABLE 3. The twisted primes. “Odd twisted weight > 0 ”: curves with $E[\ell]$ irreducible, not CM, having an odd prime of type II, II* ($\ell = 3$) or I_n^* with $\ell \mid n$. “Twisted prime 2”: curves where the prime 2 is of such a type. The last column counts the violations of (2) with the weight of the prime 2 added to the right-hand side.

with exactly two odd primes of type II or II* and no other contribution, $9 \mid \deg \varphi_E$ always, and we tested whether the mechanism could be a congruence between f and its own quadratic twist by the character ramified at both primes, which would make f an eigenform with $a_p \equiv 0 \pmod{3}$ at all primes inert in the corresponding quadratic field: this never happens (0 of 2,502; the fraction of inert primes with $3 \mid a_p$ is 0.37, against $1/3$ for a random form). The congruent form of Section 3 is a different one, the twist of a form of lower level.

4.6. Out of table. To test the statements beyond the tables, we generated curves with prescribed local types and conductor between 500000 and $4 \cdot 10^6$ (families with a type IV or IV* prime, with two or three type II primes, with mixed types, and random curves), kept those with trivial isogeny class (so that the curve is optimal and $E[\ell]$ is irreducible for every ℓ) and not CM, and computed their modular degrees with `ellmoddegree`. Table 4 summarises them; Conjectures 1.1 and 1.3 hold for all of them (Table 1, last rows).

family	curves	largest conductor	median seconds per degree
II,II	98	3,968,800	0.4
II,II,II	1	2,832,200	0.2
IV	334	3,988,012	1.1
IV*	104	3,994,600	4.9
IVmixed	266	3,983,504	1.3
random	90	2,710,716	32.1
all	893	3,994,600	1.5

TABLE 4. Curves outside the tables, with modular degrees computed independently.

4.7. The Eisenstein case. Among the 75,430 working-set optimal non-CM curves with a rational 3-isogeny, 67,845 have a positive bound in (1) and 2,777 a positive deficit D ; $D = 2$ occurs 22 times. No curve without a rational 3-torsion point has a deficit, whatever its 3-isogeny structure. With a rational point of order 3 and mod 3 image of Borel type, $D \leq 1$; $D = 2$ occurs

only when $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ (the split Cartan image **3Cs.1.1** in the notation of [21], seventeen curves, all of rank 0 and torsion $\mathbb{Z}/3$) or when E has a point of order 9 (five curves). The bounds of Conjecture 1.5 hold with no exception on all three ranges, and the torsion bound is attained with $D > 0$ in 2,773 cases. For $\ell = 5$ and 7 the deficits are at most 1 and occur only with a rational ℓ -torsion point (192 and 18 cases in the working set). Measured against the stronger bound (2) the deficit is not controlled by the torsion: it exceeds the bound of Conjecture 1.5 for 1,148 working-set curves at $\ell = 3$ (189 and 39 on the hold-out and the extension) and for 141 at $\ell = 5$. In the Eisenstein case the twisted primes do not contribute reliably, and Conjecture 1.5 is stated for (1) only.

5. THE PRIME 2

At $\ell = 2$ the modular degree is the subject of Watkins's conjecture $2^{\text{rank } E(\mathbb{Q})} \mid \deg \varphi_E$ [24] and of a substantial literature [9, 4, 25, 6, 10, 3], and the mod 2 representation has a rational 2-isogeny for a large fraction of curves. We record only the following. For working-set optimal curves without a rational 2-isogeny and not CM, the inequality $v_2(\deg \varphi_E) \geq \sum_{q \parallel N} v_2(n_q) + \#\{q : \text{type III, III}^*, \text{I}_0^* \text{ or } \text{I}_n^*\}$ holds with no exception on all three ranges (964,432, 325,510 and 313,741 curves), while counting I_0^* with weight 2 = $v_2(\#\Phi_q^{\text{geom}})$ fails (123, 26 and 20 exceptions). The isolated-configuration minima suggest that types III, III* and I_n^* contribute 3 each and types II, IV and I_0^* contribute 1 each, in line with the theorem of Calegari and Emerton that a curve of odd modular degree has at most two odd bad primes and a rational 2-torsion point or prime conductor [4]. We have not attempted to reconcile the exponents with the results of [9, 6, 10, 3].

6. OPEN QUESTIONS

- (1) Prove Conjecture 1.1 at $\ell = 3$ for multiplicative primes, and remove the conditions $\ell \nmid N$ and $q \not\equiv \pm 1 \pmod{\ell}$ from Theorem 1.2. The data say the inequality survives both; whether the equality of Kim and Ota does is a separate question.
- (2) Prove the quantitative additive statements: a Kim–Ota type formula in which the prime q enters the level with exponent 2 and the local invariant is the unipotent ramification of Lemma 2.2 (types IV, IV*), or the unipotent ramification of the twist (types II, II*, I_n^*). For I_n^* the formula for $f \otimes \chi_q$ at level N/q is available, and the question is how the congruence ideals at levels N/q and N compare.
- (3) Explain why the prime 2 contributes nothing through its twisted types, while an I_n^* at 2 with $9 \mid n$ does contribute one (Table 2). Presumably the right local invariant at 2 is the conductor exponent of $\bar{\rho}_{E,\ell} \otimes \psi$ over the three quadratic characters ψ ramified at 2.
- (4) Identify the deficit of Conjecture 1.5 with a natural invariant of the Eisenstein ideal at ℓ , presumably through the cuspidal subgroup of $J_0(N)$ [13].

APPENDIX A. REPRODUCIBILITY

All scripts, the pinned data commit and the result tables are in the repository <https://github.com/dsr001001/mcptrial> (directory `ecmine`). The checker takes a table of optimal curves with their Kodaira symbols, Tamagawa numbers, modular degrees, isogeny degrees and torsion, and reports the violations of each statement; it runs on the full tables in minutes. A single curve can be checked against (2) at $\ell = 3$ in PARI/GP with

```
E = ellinit([0,0,0,-99,360]); R = ellglobalred(E); L = R[5]; P = R[4][,1];
d = ellmoddegree(E);
b = sum(i=1,#L, my(k=L[i][2], q=P[i]); if(k>=5, valuation(k-4,3),
    if(k<=-5, if(q>2, valuation(-k-4,3), 0),
    if(abs(k)==4 || (abs(k)==2 && q>2), 1, 0))));
```

```
print([valuation(d,3), b])
```

where the PARI codes $k = n+4$, $-(n+4)$, ± 2 , ± 4 stand for I_n , I_n^* , II/II^* , IV/IV^* . The pipeline was rerun from a clean clone and reproduced every count. Lemma 2.4(ii) was also checked numerically on 4,577 random pairs (curve, odd bad prime) with no exception (`scripts/twist_lemma_check.gp`).

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[AFFILIATION]

Email address: [email]